

# Tushar Parihast

Artificial Intelligence | Machine Learning | Computer Vision | Data Analysis

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## Profile

Final-year B.Tech student in Artificial Intelligence at Kathmandu University with hands-on experience in Machine Learning, Computer Vision, Natural Language Processing, Multimodal AI, and Data Analytics. Experienced in developing AI-powered solutions for healthcare, agriculture, and cultural preservation using Python, PyTorch, TensorFlow, and modern web technologies. Strong background in research-driven development, hackathons, and collaborative problem-solving with a passion for building scalable AI systems that create measurable real-world impact.

## Education

**Bachelor of Technology in Artificial Intelligence** 2022 – Present

Kathmandu University, Nepal

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Mining, Database Systems, Software Engineering, Data Structures & Algorithms

**Higher Secondary Education (+2 Science)** 2020 – 2022

United Academy, Nepal

## Technical Skills

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|-----------------------|------------------------------------------------------------------------------------------|
| <b>Programming</b>    | Python, C, C++, JavaScript, SQL, Typescript                                              |
| <b>AI / ML</b>        | PyTorch, TensorFlow, Scikit-Learn, OpenCV, Hugging Face                                  |
| <b>Core AI Areas</b>  | Computer Vision, Multimodal AI, NLP, Agentic AI, Deep Learning, Data Science             |
| <b>Data Analytics</b> | Pandas, NumPy, Matplotlib, Data Visualization, Data Cleaning, Feature Engineering        |
| <b>Frontend</b>       | React, React Native, Vite, Tailwind CSS                                                  |
| <b>Databases</b>      | PostgreSQL, MySQL, Supabase                                                              |
| <b>Tools</b>          | Git, GitHub, Linux, VS Code, Jupyter Notebook, Google Colab, Docker                      |
| <b>Soft Skills</b>    | Leadership, Team Coordination, Communication, Research, Public Speaking, Problem Solving |

## Projects

### Multimodal Nepalese Cultural Music Video Classification

*Stack: Python, PyTorch, Computer Vision, Audio Processing, Multimodal AI*

- Developed a multimodal AI system for classification and preservation of Nepalese cultural music videos from diverse indigenous communities.
- Collected and curated multimedia datasets manually from online platforms and community sources.
- Addressed limited data availability through annotation strategies and structured dataset preparation.
- Researched techniques for combining audio and video model outputs in multimodal learning pipelines.
- Contributed to platform design, research activities, and application development efforts.

### Animal Image Classification

*Stack: Python, TensorFlow, Computer Vision, Custom Dataset*

- Built a multi-class classifier to identify 20+ native domestic species of Nepal.
- Assembled and curated a custom dataset representing the diverse habitats and environmental conditions.
- Performed data augmentation to achieve high classification accuracy on a fine-grained species recognition task .

## AI-Based Body Condition Scoring System for Cattle Health

Stack: Python, PyTorch, OpenCV, Computer Vision

- Developed an AI-assisted livestock assessment system for automated cattle body condition scoring.
- Designed the complete scoring workflow and contributed to model training and evaluation.
- Conducted extensive research on livestock health indicators and agricultural monitoring methodologies.
- Built user interface components and presented project findings to academic and technical audiences.
- Supported development of technology aimed at improving livestock management and farmer decision-making.

## RecallRx: AI Companion for Dementia Care

Stack: Raspberry Pi, Python, Computer Vision, AI

- Developed an AI-assisted dementia care prototype that provides memory support, reminders, and daily assistance for elderly users.
- Integrated Raspberry Pi hardware with computer vision and AI components to create an interactive healthcare solution.
- Designed user-centric interfaces focused on accessibility, usability, and caregiver support.
- Explored practical applications of AI-driven assistive technologies for improving quality of life in dementia care.

## Vet Finder Nepal

Stack: React Native, Geoapify API, Mobile Development

- Developed a location-aware mobile application that helps users discover nearby veterinary clinics and animal healthcare services.
- Integrated Geoapify APIs for real-time location search, mapping, and veterinarian information retrieval.
- Designed a responsive mobile interface to improve accessibility to veterinary resources for pet owners and livestock farmers.
- Optimized user experience through intuitive navigation and location-based service recommendations.

**AI-Powered Automatic Doctor Appointment System** Stack: Next.js, TypeScript, Tailwind CSS, Supabase, PostgreSQL, n8n, Gemini API, Machine Learning

- Developed an AI-driven healthcare appointment platform that analyzes patient symptoms using Natural Language Processing (NLP) to identify appropriate medical specializations.
- Designed intelligent doctor-matching and automated scheduling workflows that recommend available doctors and optimize appointment allocation in real time.
- Integrated Large Language Models (LLMs) with n8n automation to handle symptom intake, department classification, appointment booking, and patient communication.
- Built a responsive multi-role system (Patient, Doctor, Admin) featuring AI-assisted decision support, real-time availability management, and automated notifications.

## Achievements & Activities

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|------|------------------------------------------------|
| 2026 | Selected Participant, eSewa Hackathon          |
| 2025 | Participant, Hack for Nepal National Hackathon |
| 2025 | First Runner-Up, Aaviskar 2025 (IoT Hackathon) |
| 2024 | Volunteer, World Youth Hackfest                |
| 2023 | Best Volunteer Award, AI Crusade               |

## Research Interests

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Computer Vision • Multimodal AI • Natural Language Processing • Agentic AI • Healthcare AI • AI for Agriculture • Data Analytics • AI for Sustainable Development

## Languages

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Nepali (Native), English (Professional), Maithali (Native), Hindi (Conversational)